

Navigating Independence: A Survey of Visually Impaired People's Experiences and Needs

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Abstract

Independent navigation in unfamiliar environments remains a major challenge for blind and visually impaired individuals despite the availability of assistive technologies. This paper presents the results of a fully accessible online survey investigating navigation experiences, challenges, and technology preferences among people with visual impairments worldwide. The survey was distributed through individuals and organizations supporting visually impaired communities. Results indicate that smartphone-based applications are the most used digital navigation aids, while a substantial proportion of participants report not using any assistive navigation technology due to cost, accessibility, or usability barriers. Participants reported persistent difficulties in obstacle detection, wayfinding, and navigation in complex environments, and expressed a preference for wearable and hands-free solutions. Overall, the findings provide a user-centered overview of navigation needs and offer insights into the design and evaluation of future assistive navigation systems.

Introduction

Independent navigation in unfamiliar environments is a fundamental requirement for autonomy, safety, and social participation. For blind and visually impaired (BVI) individuals, navigating everyday spaces such as streets, public buildings, or transportation hubs often involves significant challenges related to orientation, obstacle avoidance, and wayfinding. Although a range of assistive technologies has been developed, their adoption and effectiveness in real-world settings remain limited.

Limited use of assistive technologies should not be interpreted as a lack of need, but rather as a mismatch between user needs and the accessibility, usability, and practicality of available solutions. Factors such as cost, ease of use, physical burden, environmental conditions, and regional context strongly influence whether a system is usable in everyday situations. These considerations highlight the importance of grounding assistive navigation research in empirical user experience.

To address this, the paper presents a survey of visually impaired individuals worldwide, capturing navigation experiences, challenges, and preferences related to assistive technologies. The questionnaire included closed- and open-ended questions covering background, technology use, and navigation needs, and was distributed globally through organizations and individuals supporting visually impaired communities. This work provides a user-centered overview of navigation experiences and needs, complementing existing research that often focuses on technical development. The findings aim to inform and contextualize future research on assistive navigation systems.

The remainder of this paper is organized as follows. The next section reviews related work, followed by the survey methodology. The results are then presented and discussed, followed by limitations and conclusions.

Related Work

Research on navigation for blind and visually impaired individuals has explored a wide range of environments and methodological approaches, including outdoor and indoor settings, as well as studies spanning both contexts. Prior work has employed surveys, interviews, and observational studies to investigate navigation challenges, user needs, and interaction strategies.

A substantial body of work has focused on outdoor navigation, particularly in urban and transportation-related settings [1–7]. These studies highlight challenges related to orientation, environmental awareness, and infrastructure accessibility, and emphasize the role of auditory information and spatial understanding in supporting independent mobility.

Indoor navigation has also received considerable attention, especially in unfamiliar or structured environments such as public buildings and campuses. Prior work has identified key challenges related to wayfinding, obstacle awareness, and the availability of relevant environmental information, and has emphasized the importance of user-centered design in defining effective navigation support [8–10].

Several studies have further examined navigation behavior across different contexts and the use of assistive technologies in everyday scenarios, highlighting the influence of situational factors, user preferences, and interaction strategies [11, 12]. In parallel, a large body of work has proposed and evaluated assistive navigation systems, including wearable, vision-based, and multimodal approaches, often in controlled or system-specific settings [13, 14].

Although these studies provide valuable insights, they are often constrained by specific environments, geographic regions, or particular technologies. In contrast, the present work adopts a user-centered perspective through a globally distributed and accessible questionnaire, aiming to capture navigation experiences, challenges, and preferences reported by visually impaired individuals across diverse real-world contexts.

Survey Methodology

This study investigates the challenges, needs, and experiences of visually impaired individuals in unfamiliar environments through a fully accessible online questionnaire. Accessibility was a central design requirement, and the questionnaire was tested and refined with input from visually impaired users, supported by manual evaluation using the WAVE accessibility tool. A pilot study involving visually impaired and sighted participants was conducted to identify usability issues.

The questionnaire consisted of 17 closed-ended questions (12 single-choice and 5 multiple-selection) and two open-ended questions covering personal background, navigation experiences, and preferences. Conditional branching was applied such that follow-up questions were shown only to participants who reported using digital assistive navigation technologies. Eligibility was based on self-identification as blind or visually impaired, verified through early questions on vision status, onset, and severity. The survey aimed to include participants from diverse demographic and geographic backgrounds. Participation was voluntary and based on implied consent, and no personally identifiable information was collected.

Survey responses were cleaned by removing duplicate and incomplete submissions. All questions were required, resulting in a dataset without missing values. Responses were analyzed using quantitative methods for closed-ended questions and inductive thematic analysis [21] for open-ended responses. For key proportions, 95% confidence intervals were computed using the Wilson score interval [22]. Given the exploratory nature of the study and the relatively small sample size, the analysis focuses on descriptive statistics. Inferential or multivariate analyses were not performed, as they would not provide statistically reliable conclusions for the present dataset. The survey questionnaire and supporting documentation are available in the project repository: <https://github.com/banafshebamdad/PhD-Survey-Visually-Impaired>.

Results

This section presents the results of the survey, including participant demographics, experiences with assistive navigation technologies, and stated preferences.

Participant Background

Participants represented diverse geographic regions and demographic groups, with the largest proportion from Western Europe and North America, followed by Africa and Eastern Europe, as shown in Table 1. The sample included a broad age distribution and a balanced representation of acquired and congenital visual impairment, with most participants reporting severe impairment or total blindness.

Table 1 Participant demographics and visual impairment characteristics. Values are reported as number of participants *n* (%).

Region	Africa	11 (26.19%)	Gender	Male	25 (59.52%)
	Asia-Pacific	4 (9.52%)		Female	16 (38.10%)
	Eastern Europe	9 (21.43%)		Non-binary	1 (2.38%)
	Latin America & Caribbean	1 (2.38%)	Severity	Mild	4 (9.52%)
	Western Europe & North America	17 (40.48%)		Moderate	6 (14.29%)
Age Group	Under 15	0 (0.00%)		Severe	17 (40.48%)
	15-30	9 (21.43%)		Total blindness	15 (35.71%)
	30-45	14 (33.33%)	Onset	Acquired	24 (57.14%)
	45-60	14 (33.33%)		Congenital	18 (42.86%)
	Above 60	5 (11.90%)			

Experience with Assistive Navigation Technologies

Smartphone-based applications were the most reported assistive technologies, although a substantial proportion of participants reported not using any digital navigation tools. Non-use of digital assistive navigation technologies varied across levels of visual impairment. Participants with total blindness and moderate impairment showed higher rates of non-use (approximately one-third), whereas no non-use was observed among participants with mild impairment. This suggests that individuals with more severe vision loss may face greater barriers to adopting existing technologies. Table 2 presents these results across five categories: technology type, usage settings, overall satisfaction, feedback adequacy and responsiveness (time to react). Among users, usage was primarily oriented toward outdoor or mixed environments. Satisfaction levels varied, and many participants reported insufficient feedback, particularly in environments with dense obstacles or high ambient noise. Perceived responsiveness was also inconsistent, with many users indicating limited time to react to obstacles, highlighting potential safety concerns.

Table 2 Experiences with digital assistive navigation technologies (users only, *n*=32; *n* (%), 95% CI).

Technology	Smart cane	2 (4.8%)	[1.3, 15.8]	Adequacy	Sufficient for navigation	6 (18.8%)	[8.9, 35.3]
	Smartphone apps	28 (66.7%)	[51.6, 79.0]		Insufficient indoors	4 (12.5%)	[5.0, 28.1]
	Smartwatch apps	0 (0.0%)	[0.0, 8.4]		Insufficient outdoors	8 (25.0%)	[13.3, 42.1]
	Other technologies	2 (4.8%)	[1.3, 15.8]		Insufficient with obstacles	10 (31.3%)	[18.0, 48.6]
	None	10 (23.8%)	[13.5, 38.5]		Insufficient in noise	3 (9.9%)	[3.3, 24.2]
Setting	Indoor only	2 (6.3%)	[1.7, 20.1]		Other limitations	1 (3.1%)	[0.6, 15.7]
	Outdoor only	16 (50.0%)	[33.6, 66.4]	Responsiveness	Always sufficient	2 (6.3%)	[1.7, 20.1]
	Both in- & outdoor	14 (43.8%)	[28.2, 60.7]		Usually sufficient	11 (34.4%)	[20.1, 52.6]
Satisfaction	Very satisfied	3 (9.4%)	[3.2, 24.2]		Sometimes sufficient	12 (37.5%)	[22.9, 54.7]
	Satisfied	15 (46.9%)	[30.9, 63.6]		Rarely sufficient	4 (12.5%)	[5.0, 28.1]
	Neutral	4 (12.5%)	[5.0, 28.1]		Never sufficient	2 (6.3%)	[1.7, 20.1]
	Unsatisfied	5 (15.6%)	[6.9, 31.8]		Uncertain	1 (3.1%)	[0.6, 15.7]
	Very unsatisfied	3 (9.4%)	[3.2, 24.2]				
	Uncertain	2 (6.3%)	[1.7, 20.1]				

Participants reported frequent challenges related to route finding, landmark identification, and obstacle detection. Head-level obstacles were reported most often, followed by hazards at chest, ground, and leg levels, reflecting the multi-level nature of real-world navigation risks, as presented in Table 3.

Preferences and Expectations

Preferences and expectations were examined across three dimensions: device form, feedback modality, and desired system characteristics. Preference for wearable devices was observed across all levels of visual impairment, particularly among participants with severe and total blindness. In contrast, handheld solutions were more frequently selected by participants with severe or total impairment and were rarely preferred by those with moderate or mild impairment. A similar pattern was observed across age groups, with wearable devices preferred in all age categories. Table 4 summarizes the relationship between severity of visual impairment, technology non-use, and device preferences, with preference percentages calculated within each severity group.

Table 3 Navigation challenges and collision risk by obstacle height (n (%), 95% CI; multiple selections allowed).

Challenges	Finding indoor landmarks	22 (52.9%)	[37.7, 66.6]	Obstacle Height	Head level	32 (76.2%)	[61.5, 86.5]
	Finding outdoor landmarks	24 (57.1%)	[42.2, 70.9]		Chest level	17 (40.5%)	[27.0, 55.5]
	Detecting obstacles ahead	19 (45.2%)	[31.7, 59.5]		Leg level	16 (38.1%)	[25.0, 53.2]
	Route finding	29 (69.1%)	[54.0, 80.9]		Ground-level	17 (40.5%)	[27.0, 55.5]
	Other challenges	6 (14.3%)	[6.7, 27.6]		None	1 (2.4%)	[0.4, 12.3]

Table 4 Technology non-use and device preference by impairment severity (n, %).

Severity	Non-use	Wearable	Handheld	Other
Mild	0 (0%)	3 (75%)	1 (25%)	0 (0%)
Moderate	2 (33%)	5 (83%)	0 (0%)	1 (17%)
Severe	3 (18%)	10 (59%)	6 (35%)	1 (6%)
Total blindness	5 (33%)	8 (53%)	5 (33%)	2 (13%)

In addition to device form, participants also expressed preferences regarding how navigation information should be delivered. Audio and multimodal feedback were the most preferred options, whereas tactile-only and simple auditory cues were selected less often. Among tactile options, feedback delivered to the hands and arms was preferred most frequently, as shown in Table 5. To provide a compact overview of the relationship between current technology use and user preferences, Figure 1 compares the distribution of device categories for current usage and preferred device categories. The figure shows that most participants currently rely on handheld solutions, particularly smartphones, whereas a larger proportion express a preference for wearable technologies.

Table 5 Device and feedback preferences (n (%), 95% CI; multiple selections allowed for tactile feedback location).

Form	Wearable	26 (61.9%)	[46.8, 75.0]	Tactile Location	Hands and arms	24 (57.1%)	[42.2, 70.9]
	Handheld	12 (28.6%)	[17.2, 43.6]		Wrists	16 (38.1%)	[25.0, 53.2]
	Other	4 (9.5%)	[3.8, 22.1]		Shoulders	12 (28.6%)	[17.2, 43.6]
Feedback	Audio	17 (40.5%)	[27.0, 55.5]		Forehead	6 (14.3%)	[6.7, 27.8]
	Multimodal	13 (31.0%)	[19.1, 46.0]		Feet	5 (11.9%)	[5.2, 25.0]
	Tactile	8 (19.1%)	[10.0, 33.3]		Waist	4 (9.5%)	[3.8, 22.1]
	Beep	4 (9.5%)	[3.8, 22.1]		Legs	1 (2.4%)	[0.4, 12.3]

Participants' expectations were also reflected in the route characteristics and system features they prioritized. Obstacle-free paths and access to route information were selected most often, followed by continuous guidance. In addition, participants emphasized ease of use, low cost, reduced cognitive and physical effort, and effective obstacle avoidance. Table 6 presents these route preferences and desired assistive navigation features. To facilitate comparison of the relative importance of system features, Figure 2 provides a ranked visualization of the responses summarized in Table 6.

Table 6 Route preferences and desired assistive navigation features (n (%), 95% CI; multiple selections allowed).

Route Factors	Shortest route	16 (38.1%)	[25.0, 53.2]
	Fastest travel time	15 (35.7%)	[23.0, 50.8]
	Lowest-cost path	11 (26.2%)	[15.3, 41.1]
	Obstacle-free paths	26 (61.9%)	[46.8, 75.0]
	Route information	23 (54.8%)	[39.9, 68.8]
	Continuous guidance	20 (47.6%)	[33.4, 62.3]
	Rapid rerouting	16 (38.1%)	[25.0, 53.2]
Key Features	Low mental load	22 (52.4%)	[37.7, 66.6]
	Low physical effort	20 (47.6%)	[33.4, 62.3]
	Ease of use	34 (81.0%)	[66.7, 90.0]
	Obstacle avoidance	20 (47.6%)	[33.4, 62.3]
	Low device weight	22 (52.4%)	[37.7, 66.6]
	Low cost	27 (64.3%)	[49.2, 77.0]
	Remote assistance	10 (23.8%)	[13.5, 38.5]

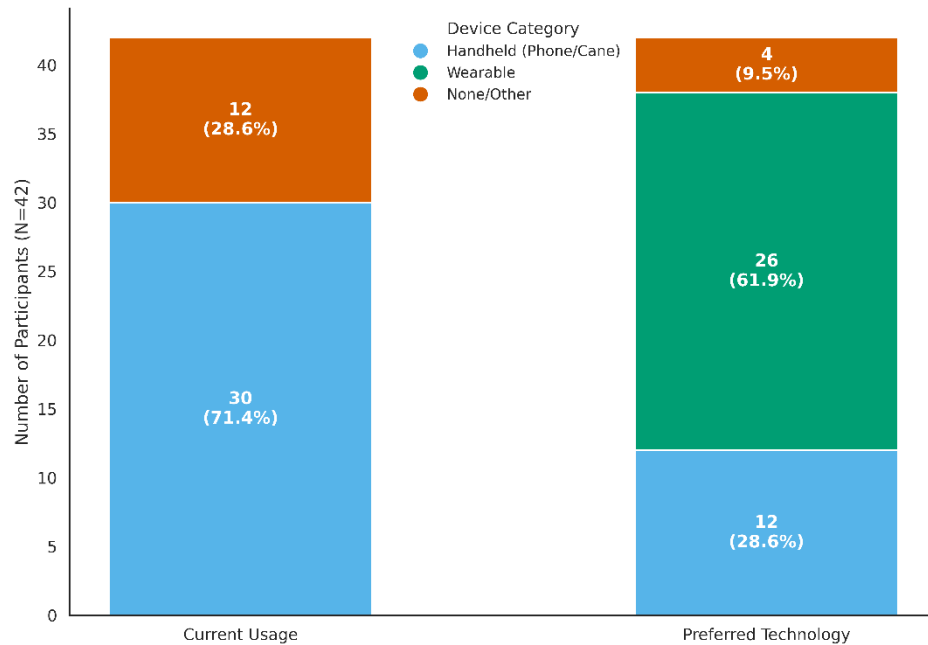


Figure 1 Comparison of current assistive technology use and preferred device categories. While most participants currently rely on handheld solutions (primarily smartphones), a majority express a preference for wearable, hands-free technologies, indicating a gap

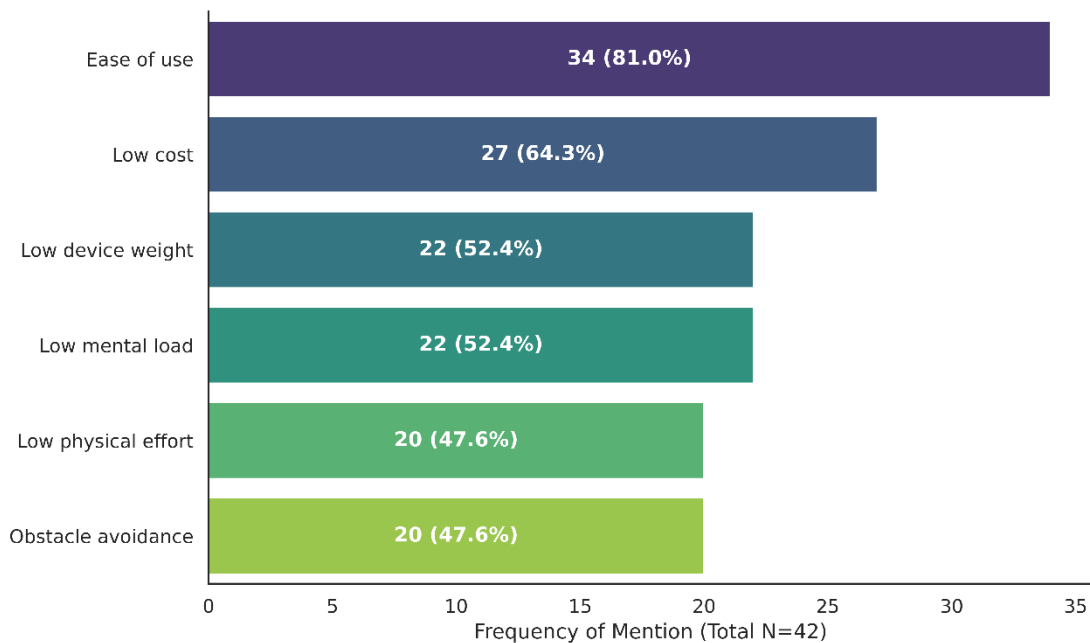


Figure 2 Importance of assistive navigation system features based on participant responses. Ease of use and low cost are the most frequently selected priorities, followed by reduced physical and cognitive effort and effective obstacle avoidance.

Insights from Open-Ended Responses

Open-ended responses emphasized simplicity, low-effort interaction, and integration with familiar, everyday devices. Participants preferred unobtrusive and discreet solutions that complement existing

mobility aids, such as canes, rather than replacing them. Reliable and actionable guidance, including obstacle-related information and clear route instructions, was frequently requested. Practical considerations, particularly affordability, accessibility across regions, and operational constraints such as offline functionality, were also highlighted.

Discussion

The findings of this study highlight persistent challenges faced by visually impaired individuals in wayfinding, obstacle avoidance, and spatial awareness. These challenges likely arise because real-world environments contain complex and multi-level obstacles that are not fully captured by current assistive technologies or traditional mobility aids. As a result, assistive navigation systems must improve their ability to detect and communicate spatial information across different obstacle levels, particularly beyond ground-level hazards. The results further indicate that outdoor environments remain especially challenging despite increased research attention on indoor navigation. This can be explained by the higher variability and unpredictability of outdoor environments, as well as the lack of standardized infrastructure, where conditions such as noise, crowd density, and layout change continuously. Therefore, assistive navigation systems should prioritize robustness and adaptability to real-world conditions rather than relying on assumptions derived from structured or controlled environments.

A clear gap is observed between the availability of assistive navigation technologies and their actual use, with many participants reporting that they do not rely on digital tools. This pattern reflects practical barriers reported by participants, including limited access, high cost, and difficulty obtaining or accessing suitable technologies, as well as environmental constraints and reliance on alternative navigation strategies. These challenges are particularly relevant for users with higher levels of impairment. In addition, reports of insufficient feedback and limited responsiveness suggest that current systems may not deliver timely and actionable information in safety-critical situations.

A further discrepancy emerges between current technology use and user preferences. While most participants rely on smartphone-based navigation in practice, this appears to reflect availability rather than optimal usability. In contrast, a majority expressed a preference for wearable, hands-free solutions, indicating a shift toward interaction modalities that reduce physical and cognitive burden. Together with the strong emphasis on ease of use and low cost, these findings indicate that non-use is driven primarily by practical barriers rather than a lack of demand.

These findings further suggest that assistive navigation systems should support multiple device form factors rather than relying on a single solution. While wearable devices enable hands-free interaction and reduced cognitive load, handheld solutions may still offer advantages in terms of control and familiarity, particularly for users with different levels of impairment and navigation strategies. Similarly, the diversity of preferred feedback modalities, including speech, multimodal, and tactile cues, reflects variability in environmental conditions and individual preferences. This indicates that no single feedback mechanism is universally effective, and that future systems should incorporate adaptable and multimodal interaction strategies that respond to both user needs and situational constraints.

Finally, participants prioritized obstacle-free routes, ease of use, and low cognitive effort over traditional efficiency metrics such as shortest path or travel time. This shift reflects the safety-critical nature of navigation for visually impaired users, where reliability and mental workload are more important than efficiency. As a result, evaluation frameworks for assistive navigation systems should move beyond purely technical metrics and incorporate user-centered criteria such as safety, cognitive load, and real-world usability.

Limitations

This study has several limitations that should be considered when interpreting the results. First, the survey relied on online recruitment, which may introduce sampling bias toward individuals with internet access and experience with digital tools. As a result, visually impaired individuals with limited access to online resources may be underrepresented. In addition, although the survey was designed to be accessible, online participation may still pose challenges for individuals who primarily rely on alternative communication methods such as Braille. The survey was conducted exclusively in English, which may have excluded non-English speakers and limited the representation of perspectives from

certain regions. Furthermore, the study was based on a relatively small subset of the broader BVI population, and the data collection period was constrained by time limitations. While efforts were made to reach a diverse group of participants within this timeframe, these factors may have influenced the overall sample size and distribution. Future studies addressing these limitations could contribute to a more comprehensive understanding of navigation challenges and needs among visually impaired individuals.

Conclusions

This paper presented the results of a fully accessible online survey examining navigation experiences, challenges, and preferences among visually impaired individuals. The findings provide a user-centered view of everyday navigation needs and highlight the diversity of experiences across users and contexts. Participants emphasized ease of use, affordability, obstacle avoidance, and access to route information, indicating that assistive navigation is evaluated not only in terms of technical performance but also in terms of safety, cognitive effort, and practical usability. These findings suggest that future systems should support flexible device form factors and interaction modalities, prioritize safety and cognitive simplicity, and provide low-effort, adaptive, and multimodal feedback. Affordability and practical accessibility should be treated as central design requirements, while evaluation should extend beyond technical metrics to include usability, reliability, and real-world effectiveness. Despite limitations related to sampling and scope, this study contributes structured evidence that can inform future assistive navigation research and development.

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